

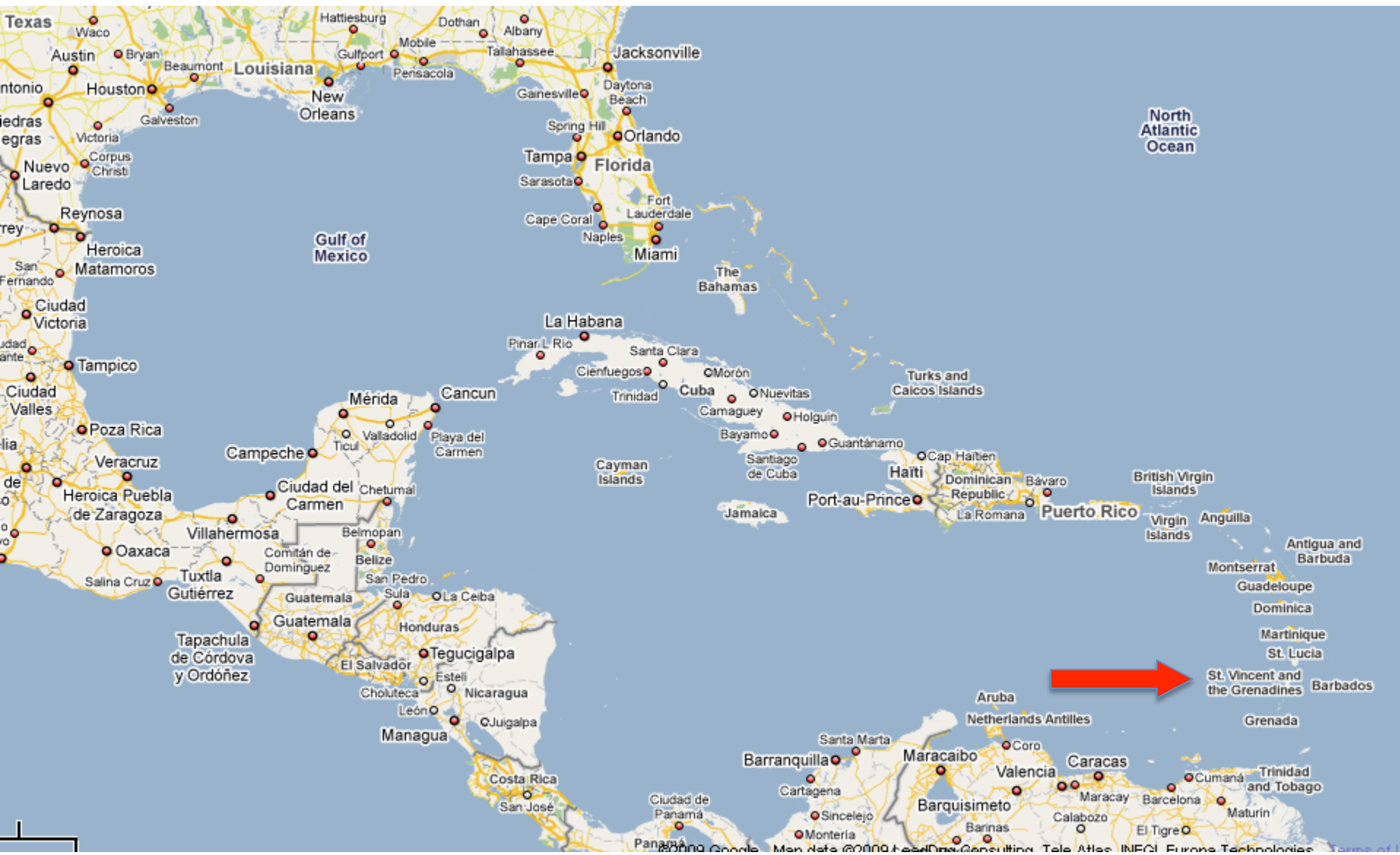
# Welcome to Computer Graphics 1

# Instructor Information

- Instructor: Dr. Reynold Bailey
- Office: GOL-3517
- Telephone: (585) 475-6181
- Office Hours: See handout or schedule on my website
- Webpage: <http://www.cs.rit.edu/~rjb/>
- My Webpage for this course:  
[http://www.cs.rit.edu/~rjb/CG1\\_20123.htm](http://www.cs.rit.edu/~rjb/CG1_20123.htm)
- Email: [rjb@cs.rit.edu](mailto:rjb@cs.rit.edu)

# A bit about me

- Born in [St. Vincent](#) in the Caribbean



# A bit about me

- Born in [St. Vincent](#) in the Caribbean
  - St. Vincent is 133 square miles
  - Lake Ontario, the smallest of the great lakes is 7,540 square miles
- Worked as a tax collector for 1 year
- Bachelors at Midwestern State University, Wichita Falls, TX
  - Double major – Math and Computer Science
- Masters and PhD in CS at Washington University in St. Louis

# A bit about me

- My research is in the field of Computer Graphics
  - Show cool video...
- My focus is on Applied Perception in Computer Graphics
  - Applying what we know about the human visual system to make better graphics devices and algorithms

# A bit about you

- Name
- Where are you from?
- Something cool about you or what you did over the break?
- Future plans – job? city? completely unsure?

# What is Computer Graphics 1

- Study of hardware and software principles for graphics
- 2D and 3D modeling and transformations
- Display algorithms
- Rendering techniques
- Graphics software packages (OpenGL)
- Implementation of fundamental algorithms and techniques.

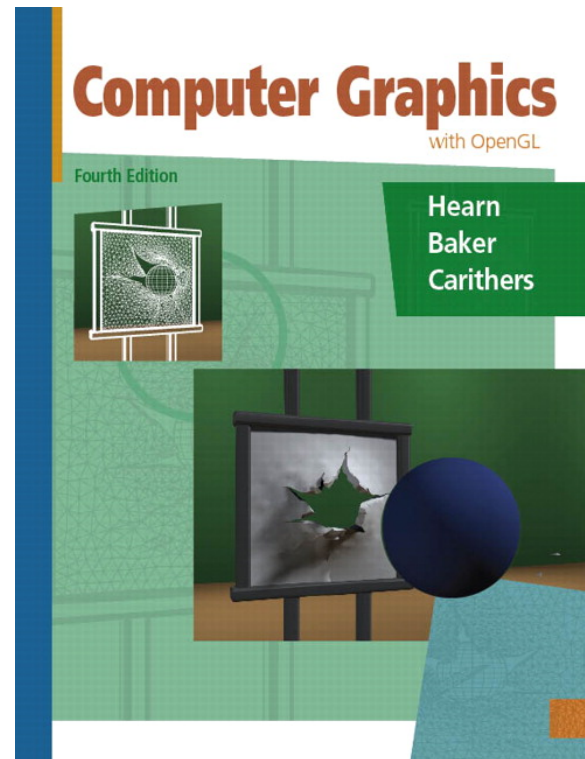


# Prerequisite

- Third year standing or permission from instructor
- Please see me as soon as possible if you have not reached this level
- This course requires a significant amount of programming
  - Be careful about taking other programming courses along with it

# Texts

- Donald Hearn, M. Pauline Baker, and Warren Carithers, Computer Graphics with OpenGL (4th Edition), Prentice-Hall, 2011, ISBN-10: 0136053580



# Texts

- Recommended:
  - OpenGL Programming Guide: The Official Guide to Learning OpenGL, Versions 3.0 and 3.1, 7th Ed.
  - OpenGL SuperBible: Comprehensive Tutorial and Reference, 5th Edition
  - OpenGL Shading Language, 3rd Edition

# Course Outcomes

- Appreciation of the history and evolution of computer graphics, both hardware and software.
- Understand/implement 2D graphics and algorithms
  - Including: line drawing, polygon filling, clipping, transformations
- Understand concepts and techniques used in 3D graphics
  - Including: viewing transformations, hierarchical modeling, color, lighting, texture mapping
- Exposure to computer graphics research areas
- Ability to use a current graphics API (OpenGL)

# Lectures

- Each class will be a series of two 50 minute lectures:
  - many will be shorter
- Around the 1 hour mark we can either take a 10 minute break or keep going (vote!)
- Questions and interaction are encouraged
  - Don't be afraid to ask questions
  - If you don't ask, I'll assume you understand what's going on

# Acknowledgements

- Material based on:
  - Textbook
- Lecture notes and assignments have been developed by several CS faculty:
  - Reynold Bailey
  - Warren Carithers
  - Joe Geigel
  - Sean Strout

# Grading Policy

|             | Undergrad | Grad |
|-------------|-----------|------|
| Exams       | 40%       | 35%  |
| Assignments | 40%       | 35%  |
| Homework    | 20%       | 20%  |
| Grad Report |           | 10%  |

# Homework

- Will be announced in class and posted on the webpage.
- Generally 1 week to complete homework assignments.
- Each homework weighted equally



# Programming Assignments

- Seven programming assignments
- Will be announced in class and posted on the webpage.
- Generally 1 weeks to complete programming assignments.
- Java or C++
- Each programming assignment weighted equally.

# Grad Report

- Each graduate student is expected to research a topic related to computer graphics and prepare a 10-15 page report.
  - Identify at least five recent papers related to the topic and summarize, compare, and discuss the work in these papers.
- GOAL: Help with grad project/thesis
- You must register topic with me by April 4th

# Exams

- Midterm: 2 hour exam
- Final: 2 hour exam: not comprehensive
- Note to grad students:
  - You are expected to have a deeper understanding of the material. Hence the undergraduate and graduate versions of the exams may be different
- Both exams will be weighted equally

# Exam Programming Option

- In lieu of the written exams:
  - Students can choose an advanced programming project for their midterm and/or their final.
  - It is expected that the code created for the weekly programming assignments will be reused and incorporated into the code for these projects.

# Policy on Late Submissions

- Late deliverables will be not be allowed
- If you foresee any problems with meeting a deadline, please see me well in advance of the deadline to work out alternate arrangement

# Submissions

- All submissions will be done using myCourses dropboxes:
  - Homework
  - Assignments
  - Grad Report Topic
  - Grad Report Final submission
  - Midterm Programming Option
  - Final Programming Option

# Tentative Schedule

[Webpage...](#)

# Academic Dishonesty

- Do Not Cheat!!!
- You are encouraged to discuss the homework and assignments freely with each other
- Do not do someone else's work for them
- Both the supplier and receiver of material are guilty of cheating
- Any help you get **must be acknowledged** in the submitted work
- Helping someone does not mean giving them your code:
  - You will get a zero for submitting someone else's code as your own
  - You will get a zero for distributing your code to others.



# Academic Dishonesty - Consequences

- First offense: Zero on the assignment
- Second offense in this or another course results in an automatic **F**
- Third offense: disciplinary action taken by RIT such as suspension or expulsion
- Links to department policy on cheating and the code of conduct for department facilities are available on the course webpage
- Please take the time to read them. Ignorance is **NOT** an acceptable excuse.

# How to do well in this class

- Show up for all lectures
- Ask questions
- Keep up with the readings listed on the webpage
- START ASSIGNMENTS EARLY
- Discuss assignments and lecture material with others
- Do Not Cheat!!!
- Work hard
- Have fun

# What's next?

- Look for an email from me:
  - Information about grader office hours
  - Username and password to access assignments/homework on webpage
- On Thursday we will watch a movie:
  - “The Story of Computer Graphics”

Questions?